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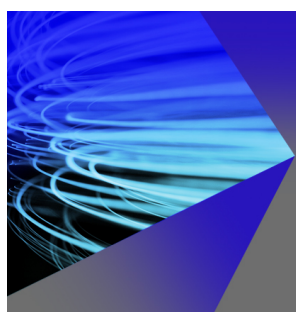
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ABSTRACT

We describe a method to experimentally determine the refractive index and thermo-optic coefficient of waveguides fabricated from low-pressure chemical vapor deposition silicon nitride over a broad wavelength range (500–1600 nm) using Mach–Zehnder interferometers. Silicon nitride refractive index Sellmeier coefficients are extracted with a weighted root mean squared deviation that is typically better than 0.04% from the visible to the near-infrared. The thermo-optic coefficient of silicon nitride is measured over a similar wavelength range and shows a significant increase at visible wavelengths compared to the value in the near-infrared. Accurate knowledge of the waveguide core and cladding refractive indices and thermo-optic coefficients in foundry-based photonic integrated circuits allows for successful first-time design of passive and active components. This method can be used with any transparent waveguide material and can be automated for process monitoring and wafer-scale verification.

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I. INTRODUCTION

Silicon nitride (SiN) is gaining widespread adoption in photonic integrated circuits (PICs) for applications that require light in the visible (400–750 nm) or I/Z-bands (750–1000 nm).^{1,2} These PICs target applications in Raman-based chemical^{3–5} and biological^{6,7} detection, augmented and virtual reality systems,⁸ and quantum technologies,^{9,10} for example. While purely passive waveguides are sufficient for some applications, other applications such as on-chip laser cavities, filter trimming, or true-time-delay require active phase or amplitude tunability. Thermo-optic phase control

is the simplest way to add active tunability to SiN waveguides and has been characterized extensively for the C-band^{11,12} and at 780¹³ and 880 nm.¹⁴ However, for active applications at shorter wavelengths,¹⁵ both the refractive index of SiN and the thermo-optic coefficient need to be accurately known. Unlike microring resonators, single-mode Mach–Zehnder interferometers (MZIs) enable broadband measurements of these parameters. Measuring the refractive index in waveguides post-fabrication is particularly important for accurate PIC component design, since thin-film based techniques, such as ellipsometry,^{16–18} do not typically capture all of the processing steps, may not have sufficient accuracy in

multilayered thin films, and are often dependent on the choice of fit model.

Unbalanced MZIs can accurately measure effective index (n_{eff}),¹⁹ which can then be used for *in situ* extraction of geometric parameters of silicon-core waveguides.^{20–25} Our recent work has shown that with four MZI spectra, one can accurately extract the four key photonic parameters for SiN waveguides: core refractive index, cladding refractive index, core thickness, and core linewidth offset (also referred to as width bias).²⁶ This method was then shown to be able to map these parameters across an entire 300-mm wafer to quantify statistical process variations.²⁷

To date, the refractive index of SiN has been reported using Mach–Zehnder interferometry only over a modest wavelength range in the O-band, and only at room temperature. Here, we extend this technique to report the refractive index of low-pressure chemical vapor deposition (LPCVD) silicon nitride across a broad optical spectrum from the visible to near-infrared wavelengths: 500–1600 nm. In addition, by performing Mach–Zehnder interferometry over a range of temperatures, we report for the first time the wavelength dependence of the thermo-optic coefficient of SiN over the same wavelength range.

II. EXPERIMENTAL SETUP AND MEASUREMENTS

Silicon nitride MZIs are fabricated using the silicon nitride passive PIC multi-project wafer (MPW) process at AIM Photonics.^{28–30} Sets of MZIs [see Fig. 1(a)] are designed for wavelengths from 500 to 1625 nm using nominally 150 nm thick (wafer TLA-VIS) or nominally 220 nm thick (wafer TLA-NIR) annealed LPCVD SiN with varying core widths and a silicon dioxide (SiO₂) cladding deposited via a plasma-enhanced tetraethyl orthosilicate (pTEOS) process.²⁸ The PICs are measured using white-light spectroscopy.^{26,31} Visible and near-infrared transmission spectra were obtained using a polarized fiber-coupled white light source (Thorlabs SLS201L or MCWHLP2) coupled to polarization-maintaining (PM) single-mode lensed fibers (Oz Optics TPMJ). The lensed fibers are used to couple light to and from the MZIs via custom edge couplers. Output light is sent to a grating-based spectrometer (Princeton Instruments SpectraPro or Wasatch Photonics WP785 and WP1064) and detected with a silicon (500–1020 nm) or InGaAs (850–1600 nm) detector array.

The transmission measurements of unbalanced MZIs can be analyzed to give accurate measurements of the effective index (n_{eff}) for both the TE₀₀ and TM₀₀ modes over a broad wavelength range.^{26,27} We use three different nominal waveguide widths for the TLA-VIS wafer: 400 nm wide (for wavelengths from 500 to 820 nm), 500 nm wide (for wavelengths from 530 to 910 nm), and 800 nm wide (for wavelengths from 600 to 1020 nm). We use four different nominal waveguide widths for the TLA-NIR wafer: 500 nm wide (for wavelengths from 680 to 1020 nm), 800 nm wide (for wavelengths from 760 to 1400 nm), 1200 nm wide (for wavelengths from 900 to 1550 nm), and 1500 nm wide (for wavelengths from 1200 to 1600 nm). These wavelength ranges are limited on the blue end by multimode propagation of the TE₀₀ mode, and on the red end by bending perturbations or loss of the TM₀₀ mode. Each waveguide width is used with sets of four MZIs that differ only in their unbalanced length. The MZI with the smallest unbalanced length is used to unambiguously identify the interference orders in the MZIs with

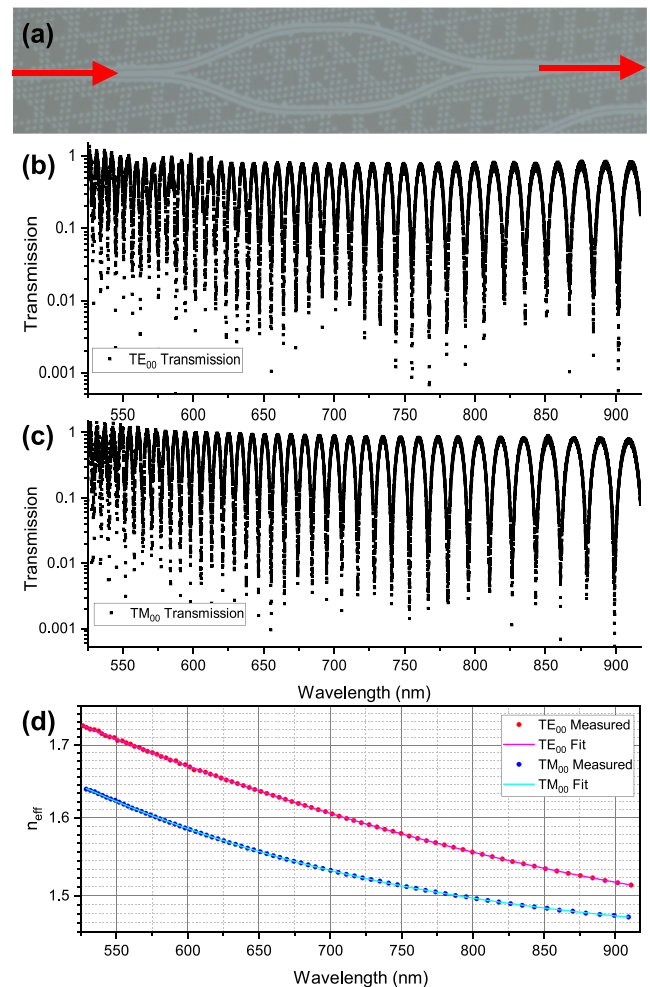


FIG. 1. (a) A microscope image of a Mach–Zehnder interferometer using 500 nm wide waveguides. (b) Measured interference spectrum for the TE₀₀ mode along with the extracted extrema. (c) Measured interference spectrum for the TM₀₀ mode along with the extracted extrema. (d) Effective index for both modes, along with a third-order polynomial fit, found from the extrema of the MZI spectra.

the larger unbalanced lengths, enabling broadband measurements of the effective index for each waveguide width.^{19,26}

Spectra can be measured at sample temperatures ranging from 20 to 125 °C using a temperature-stabilized mount. For the refractive index measurements, we measure spectra from four different locations across the wafer, to account for intra-wafer process variations in index and/or geometry.²⁷ These locations are roughly midway between the wafer center and edge. Figures 1(b) and 1(c) show example MZI spectra for the highest order MZI fabricated for the TE₀₀ and TM₀₀ modes in a 500 nm wide waveguide at 20 °C. The maxima and minima are identified using a peak/valley finding algorithm and used to determine the effective index across the whole measurement spectrum.^{26,27} The algorithm finds peaks by applying a Gaussian filter to the data before finding the local maxima. To find the valleys, the same approach is applied to the negative of the

TABLE I. Extracted geometric parameters for the two wafers under investigation.

Wafer	Thickness (nm)	Width offset (nm)
TLA-VIS	145.9 ± 0.2	-41.0 ± 0.4
TLA-NIR	216.6 ± 0.5	-51.6 ± 1.6

data. Figure 1(d) shows the resulting effective index found from this method, along with a polynomial fit, for both modes.

III. CORE GEOMETRY

Because the waveguide geometry is independent of wavelength, the core thickness and width only need to be determined at a single wavelength in order to determine the refractive index of the core and cladding over the entire measured bandwidth. Thus, we initially adopt the same approach as that used previously^{26,27} to find the core geometry for the TLA-NIR wafer and use a similar approach in the visible for the TLA-VIS wafer. Both approaches are subject to the same set of assumptions in the parameter extraction process: (i) the width offset (bias) is independent of waveguide width; (ii) the oxide cladding material parameters are uniform around the core; and (iii) the waveguide sidewall angle is vertical. Previous results have demonstrated that (i) the width offset is uniform (to within a few nm) above waveguide widths of ~400 nm; (ii) our approach gives an optical average of the cladding index; and (iii) the width offset represents a vertical midpoint value for cores with slightly (few degrees from vertical) angled sidewalls.

The parameters for the wafer TLA-NIR are found using waveguide widths of 800 and 1500 nm at a wavelength of 1300 nm. The parameters for TLA-VIS are found using waveguide widths of 400 and 800 nm at a wavelength of 700 nm. The geometry results from all four wafer locations are averaged, and the mean values are given in Table I. The errors represent one-sigma standard errors of the mean values and include both fitting errors from the effective index measurement at a given die location, and statistical variation across the four locations on the wafer.

IV. REFRACTIVE INDEX

This four-parameter extraction²⁶ can only be performed at wavelengths that correspond to spectral overlap between the ranges of the two waveguide widths used for the analysis (see Sec. II). To determine the core and cladding refractive indices over as broad an optical bandwidth as possible, the core geometry can be fixed to the values given in Table I, and the TE₀₀ and TM₀₀ effective indices can be used for each waveguide width independently.

To convert these measured effective index values to the core and cladding index, a mode-solver is used to calculate the effective indices $n_{\text{eff}(0)}^{\text{TE}}$, $n_{\text{eff}(0)}^{\text{TM}}$ for initial-guess parameters $n_{\text{core}(0)}$ and $n_{\text{clad}(0)}$. All of these parameters are understood to be wavelength-dependent. Then, if $\Delta n_{\text{core}}/n_{\text{core}}$ and $\Delta n_{\text{clad}}/n_{\text{clad}}$ are $\ll 1$, higher-order (nonlinear) dependencies between the effective indices and the refractive indices can be dropped and each n_{eff} can be written as

$$n_{\text{eff}}^{\text{TE}} = n_{\text{eff}(0)}^{\text{TE}} + \frac{\partial n_{\text{eff}}^{\text{TE}}}{\partial n_{\text{core}}} \Delta n_{\text{core}} + \frac{\partial n_{\text{eff}}^{\text{TE}}}{\partial n_{\text{clad}}} \Delta n_{\text{clad}}, \quad (1)$$

$$n_{\text{eff}}^{\text{TM}} = n_{\text{eff}(0)}^{\text{TM}} + \frac{\partial n_{\text{eff}}^{\text{TM}}}{\partial n_{\text{core}}} \Delta n_{\text{core}} + \frac{\partial n_{\text{eff}}^{\text{TM}}}{\partial n_{\text{clad}}} \Delta n_{\text{clad}}, \quad (2)$$

where the effective indices on the left-hand sides are the measured values, the partial derivatives are found from a mode solver, and the Δn are the offsets from the initial guesses that are to be solved for. For initial parameter guesses that turn out to satisfy $|n_{\text{eff}} - n_{\text{eff}(0)}| \ll 1$ (typically within a few percent), the linearity assumption of this approach is accurate and only 1 iteration is necessary. Otherwise, an iterative approach that uses a second set of initial guesses given by $n_{\text{core}(0)} + \Delta n_{\text{core}}$, etc., can be used.

To solve for the refractive indices, these equations can be rewritten as

$$\begin{pmatrix} n_{\text{eff}}^{\text{TE}} - n_{\text{eff}(0)}^{\text{TE}} \\ n_{\text{eff}}^{\text{TM}} - n_{\text{eff}(0)}^{\text{TM}} \end{pmatrix} = \begin{pmatrix} \frac{\partial n_{\text{eff}}^{\text{TE}}}{\partial n_{\text{core}}} & \frac{\partial n_{\text{eff}}^{\text{TE}}}{\partial n_{\text{clad}}} \\ \frac{\partial n_{\text{eff}}^{\text{TM}}}{\partial n_{\text{core}}} & \frac{\partial n_{\text{eff}}^{\text{TM}}}{\partial n_{\text{clad}}} \end{pmatrix} \begin{pmatrix} \Delta n_{\text{core}} \\ \Delta n_{\text{clad}} \end{pmatrix} \quad (3)$$

The Δn are found from solving Eq. (3) by inverting the matrix of partial derivatives (the Jacobian matrix), so that the refractive indices are given by

$$\begin{pmatrix} n_{\text{core}} = n_{\text{core}(0)} + \Delta n_{\text{core}} \\ n_{\text{clad}} = n_{\text{clad}(0)} + \Delta n_{\text{clad}} \end{pmatrix} \quad (4)$$

This procedure is carried out as a function of wavelength for the entire measurement spectrum of a given waveguide width and thickness. The fit errors in the measured $n_{\text{eff}}(\lambda)$ values are propagated through this solution to give errors in the refractive indices.

This process is carried out for the three waveguide widths on TLA-VIS and the four waveguide widths on TLA-NIR measured at four wafer locations, resulting in 28 refractive index datasets for each material. These spectra are plotted in Fig. 2, along with a two-term Sellmeier nonlinear least-squares fit for the SiN and separate two-term Sellmeier fits for the SiO₂ for each wafer. Also included in Fig. 2 for reference is a commonly used refractive index spectrum for LPCVD SiN found from ellipsometry³² and the refractive index of thermal SiO₂ found from prism coupling.¹⁶

The fitted Sellmeier refractive index functions are

$$n_{\text{SiN}}^2 = 1 + \frac{A_1 \lambda^2}{\lambda^2 - B_1^2} + \frac{A_2 \lambda^2}{\lambda^2 - B_2^2}, \quad (5)$$

$$n_{\text{SiO}_2}^2 = 1 + \frac{A_1 \lambda^2}{\lambda^2 - B_1^2} - C \lambda^2. \quad (6)$$

It should be noted that the SiO₂ fits are unable to independently separate the two fit parameters in the second term of a standard Sellmeier fit, so these parameters are combined in the form shown in Eq. (6). The fitted parameters from Eqs. (5) and (6) are shown in Table II.

The ~1% discrepancy between our SiN refractive index and that of Luke *et al.*³² is not presently understood, although we have also observed differences in the SiN refractive index found from MZI

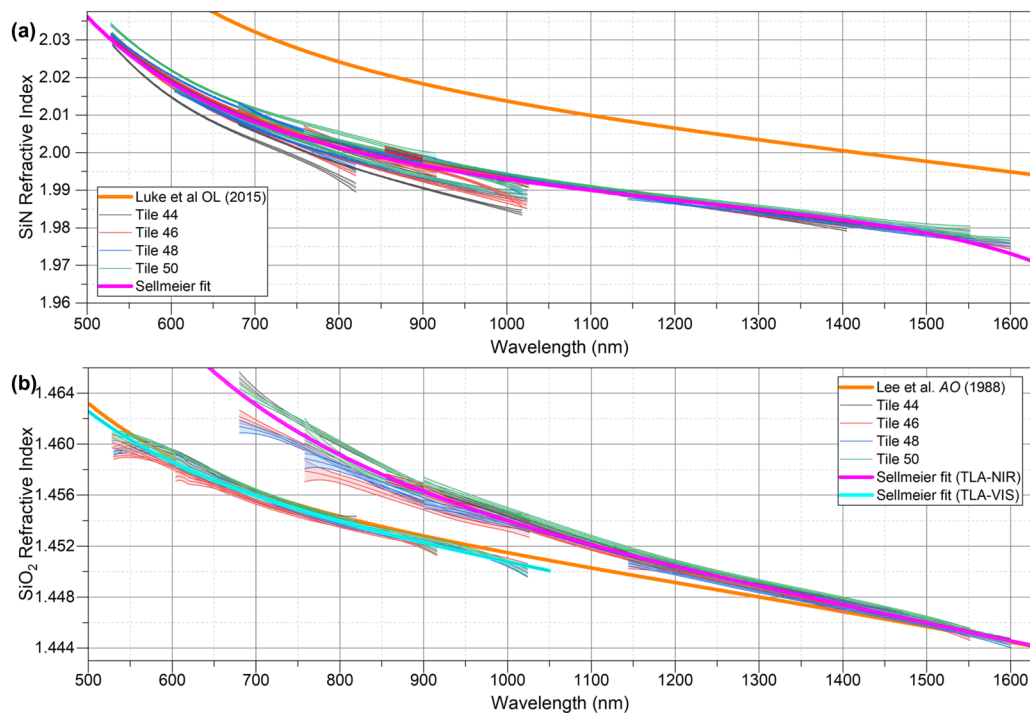


FIG. 2. (a) Measured SiN refractive index vs wavelength, and a two-term Sellmeier fit. (b) Measured SiO₂ refractive index vs wavelength, and a two-term Sellmeier fit for each of the wafers. The measurements show 1- σ errors as shaded regions.

TABLE II. Sellmeier parameters found from the fits to the data in Fig. 2.

Material	A ₁	B ₁ (μm)	A ₂	B ₂ (μm)
SiN	2.9277 ± 0.0006	0.1321 ± 0.0003	0.0200 ± 0.0016	1.8703 ± 0.0241

Material	A ₁	B ₁ (μm)	C (μm ⁻²)
SiO ₂ -NIR	1.1040 ± 0.0002	0.1331 ± 0.0005	0.0097 ± 0.0001
SiO ₂ -VIS	1.1094 ± 0.0004	0.0848 ± 0.0006	0.0127 ± 0.0004

analysis and ellipsometry performed on the same film prior to processing into waveguides. It is possible that different SiN LPCVD process conditions, such as anneal temperature and duration or furnace temperature, can alter the SiN index. The lowest-order SiN Sellmeier absorption strength (A_1) and resonance wavelength (B_1) agree well with values reported by Bääk³³

The agreement between the measured SiO_2 index and that of Lee *et al.*¹⁶ for thermal SiO_2 is very good. Previous reports of the lowest-order Sellmeier resonance wavelength, B_1 , for SiO_2 ^{33,34} are similar to those found here. Since the top cladding is deposited and not thermal in the measured waveguides, some difference between the index of thermal SiO_2 and our value is to be expected. The discrepancy between the measured SiO_2 index of the two wafers, which is between 0.5% and 0.2% between 700 and 1000 nm, could be due to slightly different process conditions or wafer-to-wafer variability. Stress differences of over 100 MPa have been reported in

different thicknesses of otherwise identical LPCVD SiN films,¹⁸ which can then differentially strain the deposited SiO₂ above.

To estimate the goodness of the Sellmeier fits for the refractive indices, we can calculate the weighted root mean square deviation (WRMSD), given by

$$\text{WRMSD} = \sqrt{\frac{1}{N} \frac{\sum_i^N (n_i^{\text{meas}} - n_i^{\text{fit}})^2 / \sigma_i^2}{\sum_i^N 1 / \sigma_i^2}}. \quad (7)$$

The weights are $1/\sigma^2$, where σ is the standard error of the refractive index, n_i^{fit} is the Sellmeier fit, i corresponds to a specific waveguide width, thickness, and wafer position, and N is the total number of these waveguides used for the fit. N ranges between 4 and 20, depending on the wavelength. We find that the WRMSD for n_{SiN} is typically below 0.0007 (0.04%), although it increases at wavelengths

above 1550 nm due to poor modal confinement in the SiN core. The WRMSD for $n_{\text{SiO}_2\text{-VIS}}$ and $n_{\text{SiO}_2\text{-NIR}}$ is typically below 0.0002 (0.01%), with an increase at wavelengths below 600 nm, due to poor modal overlap with the cladding at shorter wavelengths.

V. EXPERIMENTAL VALIDATION

To experimentally evaluate the accuracy of our extracted material indices and waveguide geometries, we performed two distinct validation exercises. First, we compared the measured effective indices [see Fig. 1(d)] in an MZI built from 500 nm wide waveguides to the simulated effective indices from a mode-solver using the extracted parameters. We note here that MZIs using 500 nm wide waveguides were not included in the geometry extraction process. The results (for several wavelengths) are plotted in Fig. 3(a). Furthermore, we repeated the same modal simulation with the SiN refractive index and waveguide geometries provided by AIM Photonics ellipsometry and critical-dimension scanning electron microscopy (CD-SEM). As shown in Fig. 3(a), the simulated effective indices based on our extracted parameters more accurately predict the measured effective indices than the parameters extracted from ellipsometry and CD-SEM. This disagreement stems partially from the fact that the ellipsometry methods used by AIM Photonics are optimized for thickness measurements rather than high-accuracy refractive index extraction.

The second validation process involves a measured and simulated mode degeneracy.³¹ In a rectangular waveguide that supports the TE_{10} mode, there is a wavelength at which the TM_{00} and the TE_{10} modes are hybridized, which allows incoming light to excite a superposition of both modes. The hybridized modes then beat as they propagate down the waveguide due to their different effective indices. This causes the mode overlap between the superposition of waveguide modes and the fundamental mode of the output fiber to vary rapidly with wavelength near the degeneracy, resulting in minima in the measured transmission spectrum due to reduced output coupling. One such dip is shown in Fig. 3(b), for a nominally 900 nm

waveguide. The wavelength of this degeneracy can be modeled in a mode-solver. We found the simulated crossing wavelength for each waveguide width for which we had measured data, first with our extracted parameters, and then with the foundry metrology parameters. As shown in Fig. 3(b), the crossings for the 900 nm wide waveguide simulated using the extracted parameters agree much more closely with the measurement than those found from the foundry metrology parameters. Note that these measurements are made using waveguide widths (varying from 800 to 1400 nm) that are significantly different from those used in the parameter extraction process, representing an independent check of our extracted parameters. The difference between the simulated and measured mode-crossing wavelength for a number of waveguide widths are plotted in Fig. 3(c). Although our extracted parameters do not predict the mode degeneracy exactly, they give results that are approximately 7x more accurate than those predicted by ellipsometry and CD-SEM data.

VI. THERMO-OPTIC COEFFICIENT

In order to measure the thermo-optic coefficient of SiN and SiO_2 , we mount the sample containing our MZIs on a ceramic heater with closed-loop temperature control. Light is coupled through the MZI as described in Sec. II, and the transmission is measured in a fixed grating spectrometer (Wasatch Photonics WP-VIS). Figure 4(a) shows the normalized transmission near a single TE_{00} maximum of a TLA-VIS (500 nm wide waveguide) MZI for several heater temperatures. By tracking the peak wavelength vs temperature with a linear fit as shown in Fig. 4(b), we can extract the derivative of peak wavelength with respect to temperature ($d\lambda_{\text{TE}}/dT$). We can then convert $d\lambda_{\text{TE}}/dT$ to the partial derivative of the TE_{00} effective index with respect to temperature ($\partial n_{\text{eff}}^{\text{TE}}/\partial T$), as:¹²

$$\frac{\partial n_{\text{eff}}^{\text{TE}}}{\partial T} = \frac{n_g}{\lambda_{\text{TE}}} \frac{d\lambda_{\text{TE}}}{dT} - n_{\text{eff}}^{\text{TE}} \alpha_{\text{Si}}, \quad (8)$$

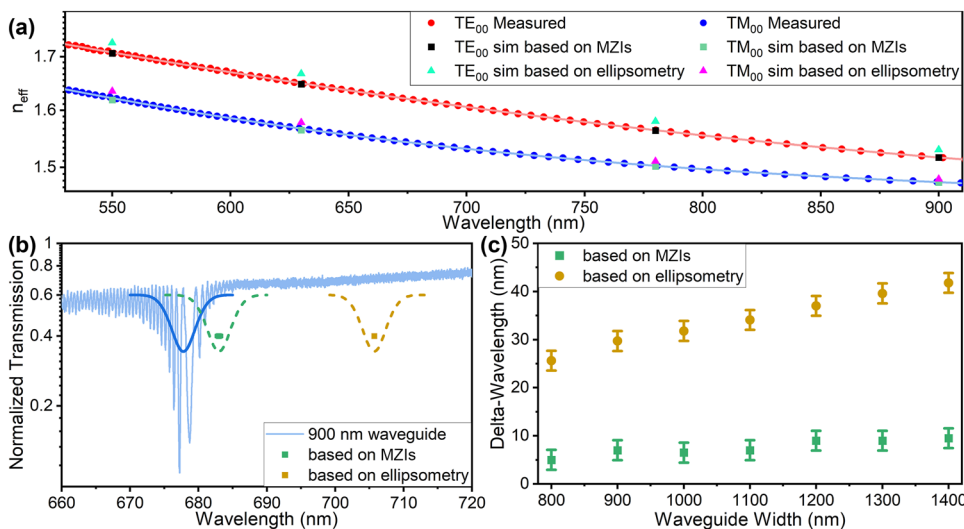


FIG. 3. (a) Measured (from MZIs) and simulated effective indices for 500 nm wide waveguides vs wavelength. The simulations are based on geometric parameters extracted in this work or from foundry metrology. (b) An example of a measured transmission minimum corresponding to a $\text{TM}_{00}/\text{TE}_{10}$ mode-crossing in a 900 nm wide waveguide. Also plotted are a fit to the measurement (dark blue solid), a simulated dip using parameters extracted in this work (green dashed), and a simulated dip using parameters extracted from foundry metrology (gold dashed). (c) A plot of the wavelength difference between the simulated mode-crossings and the measured mode crossings for a number of different waveguide widths.

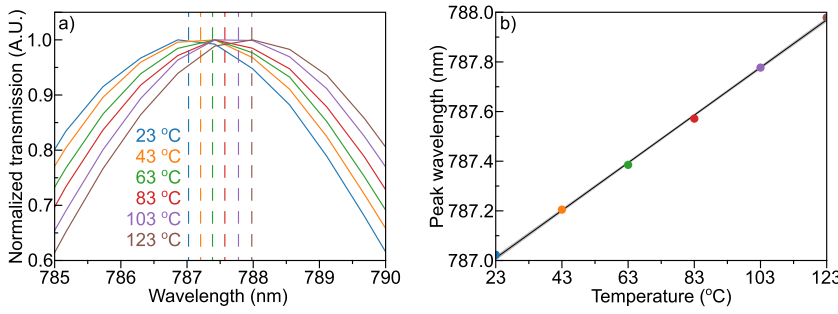


FIG. 4. (a) Normalized transmission near a single TE_{00} transmission maxima of a TLA-VIS (500 nm wide waveguide) MZI for several heater temperatures. The dashed vertical lines indicate the peak wavelengths. The spectrum at each temperature is normalized to its own maximum. (b) The peak wavelength from (a) as a function of heater temperature with a linear fit superimposed. The slope of this fit equals $d\lambda_{TE}/dT$.

where n_g is the group index of the optical mode and α_{Si} is the thermal expansion coefficient of Si, which is included to account for lengthening of the MZI as a result of expansion of the Si substrate. The expansion coefficients of the SiO_2 and SiN are expected to have a negligible effect in comparison and are, therefore, neglected. The value of α_{Si} as a function of temperature was taken from Okada and Tokumaru.³⁵ We repeat this process for the TM_{00} mode to obtain $\partial n_{eff}^{TM}/\partial T$.

To obtain the thermo-optic coefficients of SiO_2 ($\partial n_{SiO_2}/\partial T$) and SiN ($\partial n_{SiN}/\partial T$), we use¹²

$$\begin{pmatrix} \frac{\partial n_{eff}^{TE}}{\partial T} \\ \frac{\partial n_{eff}^{TM}}{\partial T} \end{pmatrix} = \begin{pmatrix} \frac{\partial n_{eff}^{TE}}{\partial n_{core}} & \frac{\partial n_{eff}^{TE}}{\partial n_{clad}} \\ \frac{\partial n_{eff}^{TM}}{\partial n_{core}} & \frac{\partial n_{eff}^{TM}}{\partial n_{clad}} \end{pmatrix} \begin{pmatrix} \frac{\partial n_{SiN}}{\partial T} \\ \frac{\partial n_{SiO_2}}{\partial T} \end{pmatrix}, \quad (9)$$

where the 2×2 Jacobian matrix above is calculated via a mode solver as described in Sec. IV. We can then solve for the thermo-optic coefficients by inverting the following matrix:

$$\begin{pmatrix} \frac{\partial n_{SiN}}{\partial T} \\ \frac{\partial n_{SiO_2}}{\partial T} \end{pmatrix} = \begin{pmatrix} \frac{\partial n_{eff}^{TE}}{\partial n_{core}} & \frac{\partial n_{eff}^{TE}}{\partial n_{clad}} \\ \frac{\partial n_{eff}^{TM}}{\partial n_{core}} & \frac{\partial n_{eff}^{TM}}{\partial n_{clad}} \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial n_{eff}^{TE}}{\partial T} \\ \frac{\partial n_{eff}^{TM}}{\partial T} \end{pmatrix} \quad (10)$$

This process is repeated for each peak/valley within the transmission spectrum, and the results are shown in Fig. 5 for six different waveguide geometries. To obtain a functional form for the fits shown in Fig. 5, we refer to the method described by Ghosh, which gives³⁶

$$n^2 - 1 = \frac{Fs\lambda_0^2(\Delta T)}{\lambda_p^2(\Delta T)} \frac{\lambda^2}{\lambda^2 - \lambda_0^2(\Delta T)}, \quad (11)$$

$$2n \frac{\partial n}{\partial \Delta T} = GR + HR^2, \quad (12)$$

$$R \equiv \frac{\lambda^2}{\lambda^2 - \lambda_0^2(\Delta T)}, \quad (13)$$

$$G \equiv -\frac{2Fs\lambda_0^2(\Delta T)}{\lambda_p^3(\Delta T)} \frac{d\lambda_p}{dT}, \quad (14)$$

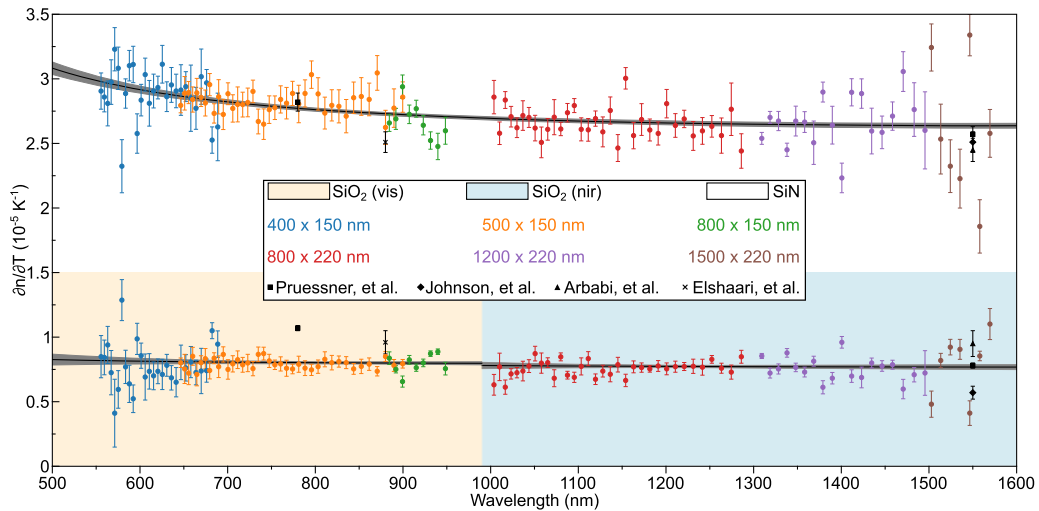


FIG. 5. Extracted material thermo-optic coefficients for SiO_2 (lower regions) and SiN (upper region) with Ghosh fit superimposed. Previously reported values from Pruessner *et al.*,¹³ Johnson *et al.*,¹² Arbabi and Goddard,¹¹ and Elshaari *et al.*¹⁴ are included for comparison. The color of the data corresponds to the waveguide dimension (width $\times$ height). All error bars correspond to one standard deviation.

$$H \equiv \frac{2F_s\lambda_0(\Delta T)}{\lambda_p^2(\Delta T)} \frac{d\lambda_0}{dT}, \quad (15)$$

where $\Delta T = 0$ at room temperature, λ_p is the material plasma wavelength, λ_0 is the material absorption resonance wavelength (the band edge), λ is the optical wavelength of interest, F_s is the effective absorption strength, and all derivatives are evaluated at $\Delta T = 0$.

By comparing Eq. (11) to a single-term Sellmeier equation, we can obtain the temperature dependence of the Sellmeier coefficients in terms of G and H , as follows:

$$(n^2 - 1)_{\text{Sellmeier}} = \frac{A_1(\Delta T)\lambda^2}{\lambda^2 - B_1^2(\Delta T)}, \quad (16)$$

$$\frac{A_1(\Delta T)\lambda^2}{\lambda^2 - B_1^2(\Delta T)} = \frac{F_s\lambda_0^2(\Delta T)}{\lambda_p^2(\Delta T)} \frac{\lambda^2}{\lambda^2 - \lambda_0^2(\Delta T)}, \quad (17)$$

$$A_1 = \frac{F_s\lambda_0^2(\Delta T)}{\lambda_p^2(\Delta T)}, \quad (18)$$

$$B_1 = \lambda_0. \quad (19)$$

We then define

$$a \equiv \frac{dA_1}{dT}, \quad (20)$$

$$b \equiv \frac{dB_1}{dT} = \frac{d\lambda_0}{dT}, \quad (21)$$

and combine Eqs. (14), (15), (18), (20) and (21) to obtain

$$a = G + H, \quad (22)$$

$$b = \frac{B_1 H}{2A_1} = \frac{H\lambda_p^2}{2F_s\lambda_0}. \quad (23)$$

The parameters G and H in Eq. (12) are fit to the data in Fig. 5, and the corresponding values of a and b are calculated. We attribute the increase in data variability at wavelengths above 1500 nm to poor modal confinement in the SiN core. For SiN, the value of $\lambda_0 = B_1$ is taken from Table II and is used to fit the entire span of data for wavelengths between 500 and 1600 nm. For SiO₂, $\lambda_0 = B_1$ from TLA-VIS is used to fit the data from 150 nm thick waveguides (wavelengths between 500 and 990 nm), while the value from TLA-NIR is used to fit the data from 220 nm thick waveguides (wavelengths between 1000 and 1600 nm). We also restrict the value of G to be negative and

H to be positive for reasons described by Ghosh.³⁶ The extracted values for G and H are shown in Table III along with the corresponding values of a and b .

From Ghosh,³⁶ We expect values of H with magnitude larger than the magnitude of G , meaning that the fractional shift in λ_0 is expected to be larger in magnitude than the fractional shift in λ_p , as expressed by

$$\frac{1}{\lambda_p} \frac{d\lambda_p}{dT} = \frac{-G}{2A_1}, \quad (24)$$

$$\frac{1}{\lambda_0} \frac{d\lambda_0}{dT} = \frac{H}{2A_1}. \quad (25)$$

For SiN, the values of G and H extracted from the fit support this claim, with G being about half the value of H . For SiO₂, the values of G and H are difficult to separate precisely due to the wavelength-independence of the measured thermo-optic coefficient. Even so, the values of G and H obtained from the fits agree well with those reported by Ghosh.³⁶ In addition, the predicted values of thermo-optic coefficient at $\lambda = 1550$ nm agree well with values reported by Pruessner *et al.*,¹³ Johnson *et al.*,¹² and Arbabi and Goddard¹¹ for both SiN and SiO₂.

Although an increase in the thermo-optic coefficient of SiN at 780 nm compared to 1550 nm has previously been reported,¹³ the measured increase in the SiN thermo-optic coefficient at visible wavelengths (18% at 550 nm compared to 1550 nm) in the present work is, to our knowledge, the first report of this wavelength-dependence measured continuously from visible wavelengths (i.e., below 780 nm) through the near-IR (1600 nm). Such an increase is important for designs of thermo-optic devices in the visible or I/Z-bands in SiN PICs. In addition, the Ghosh model of the thermo-optic effect suggests that the dominant contribution to the SiN thermo-optic coefficient is the band edge shift, even though the measurements are made at wavelengths that are hundreds of nanometers away from the band edge. Using Eqs. (24) and (25), we can rewrite Eq. (12) as

$$2n \frac{\partial n}{\partial \Delta T} = -2A_1 \left(\frac{1}{\lambda_p} \frac{d\lambda_p}{dT} R - \frac{1}{\lambda_0} \frac{d\lambda_0}{dT} R^2 \right). \quad (26)$$

Because $R \approx 1$ in the wavelength range from 0.5 to 1.6 μm and because the fractional shift in band edge with temperature extracted from the Ghosh model is approximately twice that of the plasma wavelength, we can see that the majority of the thermo-optic effect stems from the band edge shift.

The fit to the SiN thermo-optic data gives a room-temperature band edge temperature dependence of $4.4 \pm 0.8 \times 10^{-6} \mu\text{m/K}$

TABLE III. Table of extracted parameters for Eq. (12) when fit to the data in Fig. 5, along with corresponding a and b values. The errors reported correspond to one standard deviation.

Material	G (K^{-1})	H (K^{-1})	a (K^{-1})	b ($\mu\text{m K}^{-1}$)
SiN	$-9.49 \pm 3.50 \times 10^{-5}$	$19.69 \pm 3.42 \times 10^{-5}$	$10.20 \pm 0.09 \times 10^{-5}$	$4.4 \pm 0.8 \times 10^{-6}$
SiO ₂ -NIR	$0 \pm 10.47 \times 10^{-5}$	$2.19 \pm 10.34 \times 10^{-5}$	$2.19 \pm 0.13 \times 10^{-5}$	$1.3 \pm 6.2 \times 10^{-6}$
SiO ₂ -VIS	$0 \pm 6.54 \times 10^{-5}$	$2.28 \pm 6.46 \times 10^{-5}$	$2.28 \pm 0.08 \times 10^{-5}$	$0.9 \pm 2.5 \times 10^{-6}$

($-310 \pm 60 \mu\text{eV/K}$), which is the first report, to our knowledge, of this material parameter.

Under the assumption that the plasma wavelength is proportional to volume and that the SiN expands isotropically, we can define an effective linear thermal expansion coefficient (α_{eff}) for the SiN waveguide, as follows:

$$\alpha_{\text{eff}} \equiv \frac{1}{3\lambda_p} \frac{d\lambda_p}{dT} = -\frac{G}{6A_1}. \quad (27)$$

We note here that the SiN waveguide expansion is constrained by the expansion of the surrounding materials and is likely anisotropic. As a result, α_{eff} is not a direct measure of the bulk linear SiN thermal expansion coefficient. Equation (27) yields a value of $\alpha_{\text{eff}} = 5.4 \pm 2 \times 10^{-6} \text{ K}^{-1}$, which agrees, within error, with reported literature values of the bulk linear SiN thermal expansion coefficient.^{37,38}

VII. CONCLUSIONS AND FUTURE WORK

By using broadband MZIs to measure effective index in single-mode waveguides, we have measured the refractive index and thermo-optic coefficient of LPCVD SiN at wavelengths from 500 to 1600 nm. This refractive index measurement includes an estimate of measurement error and intra-wafer variability, the weighted root mean square deviation, that is typically below 0.0007 (0.04%). The material refractive indices and waveguide geometries we extract more closely predict the measured modal degeneracy wavelengths than the same parameters obtained via foundry metrology. This is also, to our knowledge, the first report of the thermo-optic coefficient of SiN at wavelengths in the visible. We observe a significant increase in the thermo-optic coefficient at visible wavelengths, compared to the near-infrared, which is important to properly design SiN thermal switches and modulators in photonic integrated circuits.

The method used here can be applied to any transparent waveguide material. By implementing it either as an inline or back-end process, wafer-scale variability can be measured and tracked, both to give designers statistical information for robust component design and also to give foundries information about tool and process performance. As more materials are added to standard photonic fabrication processes, accurate knowledge of these material's optical properties is essential to successful circuit design.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Steven T. Lipkowitz: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jordan N. Butt:** Data curation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Zachary T. Zern:** Data curation (equal); Formal analysis (equal). **Kyle J. Walsh:** Data curation (equal); Writing – review & editing (equal). **Nathan F. Tyndall:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Marcel W. Pruessner:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jacob N. Bouchard:** Formal analysis (equal); Writing – review & editing (equal). **Scott A. Holmstrom:** Conceptualization (equal); Formal analysis (equal); Methodology (equal); Writing – review & editing (equal). **Benjamin L. Miller:** Supervision (equal); Writing – review & editing (equal). **Alin Antohe:** Investigation (equal); Resources (equal). **Todd H. Stievater:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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